



WELCOME

TO My PRESENTATION



Advancements in Automated Brain Tumor Detection Using Deep Learning on MRI Imagery

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01

Introduction and Background

Topics Covered:

- Global Impact of Brain Tumors
- Challenges in MRI Diagnosis
- Access to Neuroimaging in Low-Resource Settings
- Advancements in AI and Deep Learning for Detection
- Challenges in Model Generalization and Real-Time Application

Introduction

- Brain Tumor Significance**
- MRI for Diagnosis**
- Human Interpretation Errors**
- Role of Artificial Intelligence**
- Transfer Learning Advantage**

- Existing Diagnostic Challenges**
- Dataset and Model Limitations**
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Background

- **Increased Demand for Faster Diagnosis**
- **Complexity of Tumor Types**
- **Limitations of Traditional Imaging**
- **Potential of Deep Learning**
- **Need for Robust Models.**

02

Research Objectives

Topics Covered:

- Research Objectives
 - Research Questions
-

Research Objectives

- **Develop an AI framework** for automatic brain tumor classification using transfer learning on Kaggle and hospital datasets.
- **Compare CNNs** (VGG16, EfficientNetB0, MobileNetV2) for classification performance.
- **Fine-tune CNN models** with limited data to improve accuracy.
- **Apply data augmentation** (brightness/contrast 0.8–1.2) for robustness and artifact reduction.
- **Achieve >95% accuracy** and <50ms latency for real-time detection.
- **Evaluate MobileNetV2** for real-time, edge-deployable use in low-resource hospitals.
- **Increase early detection by 25%** and provide open-source tools by October 2025.

Research Questions

- How does AI improve brain tumor diagnosis?
- Which CNN model is best for classification?
- What data augmentation techniques are used?
- How effective is MobileNetV2 in low-resource settings?
- What impact does the system have on early detection?

03

Methodology and model analysis

Topics Covered:

- Methodology
- Model analysis

Methodology

- **Dataset Collection**
- **Data Preprocessing**
- **Model Architecture**
- **Model Training**
- **Evaluation**
- **Model Deployment (Optional / Future Work)**

Model analysis

VGG16:

- Accuracy: **95%**, F1-score: **0.95**, AUC-ROC: **0.96–0.99**
- Shows good convergence
- **48 glioma misclassified** as meningioma due to edema overlap.
- Inference time: **~80ms**, parameters: **15.8M** — accurate but heavy for real-time use.

EfficientNetB0:

- Accuracy: **~27%**, F1 unstable, AUC-ROC: **0.50**
- Failed to learn (loss 1.39–1.79), poor generalization.
- Parameters: **20.3M**, inference: **~60ms** — inefficient and unsuitable without major retraining.

Model analysis

MobileNetV2:

- Accuracy: **98.5%**, F1-score: **0.98**, AUC-ROC: ≥ 0.99
- Strong convergence (loss $\downarrow$ 0.69 $\rightarrow$ 0.038), only **16 misclassifications**.
- Inference time: **<50ms**, parameters: **6.1M** — fast, accurate, and ideal for real-time or low-resource deployment.

Best Performer: MobileNetV2 — highest accuracy, fastest speed, and best balance for clinical use.

04

Implementation and Logic

Topics Covered:

- Dataset overview and samples
- Implementation and Logic

Dataset overview and samples

- **Total:** 7,143 labeled MRI images (T1-weighted).
- **Sources:** Kaggle dataset + hospitals in Bangladesh (Ibn Sina, DMC, Comilla MC).
- **Classes:**
 - Glioma (22.6%)
 - Meningioma (24.0%)
 - Pituitary (26.0%)
 - No-tumor (27.3%)
- **Split:** 83% (train) = 5,832 | 17% (test) = 1,311.
- **Challenge:** Slight class imbalance and adult-focused images (age 30–60).

Implementation and Logic

Tumor Detection Process:

- **Step 1:** Input MRI scan is preprocessed (resized, normalized).
- **Step 2:** Early layers of the model (frozen) extract basic features from the image.
- **Step 3:** Last 20 layers are unfrozen, allowing the model to focus on tumor-specific features, refining its ability to detect the type of tumor.
- **Step 4:** Data augmentation (brightness/contrast changes) simulates real-world imaging conditions, improving model generalization.
- **Step 5:** Final output layer classifies the image into one of the four tumor categories (glioma, meningioma, pituitary adenoma, or no-tumor) using a **Softmax activation** function.

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Result Analysis and Limitations

Topics covered:

- Result Analysis
- Performance Analysis
- Model Selection
- Pretrained Model Output
- Limitation

Result Analysis

- **MobileNetV2** performed best — very fast and highly accurate.
- **VGG16** good but heavy; slower inference.
- **EfficientNetB0** showed instability and poor learning.

Performance analysis

MobileNetV2 Advantages:

- Lightweight
- Fast inference
- Strong generalization (98.5% accuracy).

VGG16:

- Good texture recognition but high computational cost

EfficientNetB0:

- High theoretical potential, but unstable on this dataset.

Model Selection

Model	Test Accuracy (%)	Macro F1-Score	Average AUC-ROC	Training Accuracy	Training Loss	Inference Time (ms)	Misclassifications (Key Example)	Parameter Count (M)
VGG16	95.0	0.95	>0.99	96.9	0.089	~80	48 gliomas as meningiomas	15.8
EfficientNet B0	~27.0	N/A (random)	0.50	26.8	1.39	~60	Uniform across classes	20.3
MobileNetV2	98.5	0.98	≥ 0.99	98.5	0.038	<50	16 meningiomas as pituitary	6.1

Pretrained Model Output

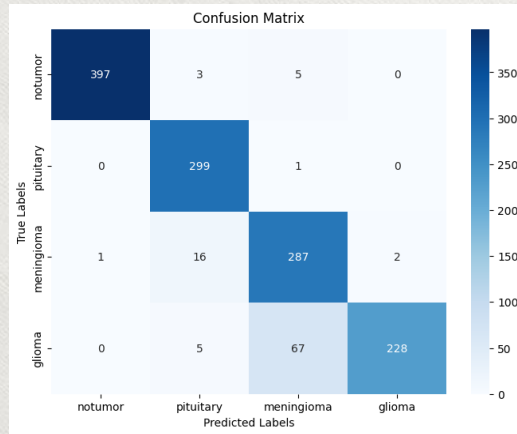


Fig: Confusion matrix MobileNetV2

Classification Report (Deeper CNN):

	precision	recall	f1-score	support
0	0.97	0.98	0.98	124
1	0.98	0.97	0.97	116
accuracy			0.97	240
macro avg	0.98	0.97	0.97	240
weighted avg	0.98	0.97	0.97	240

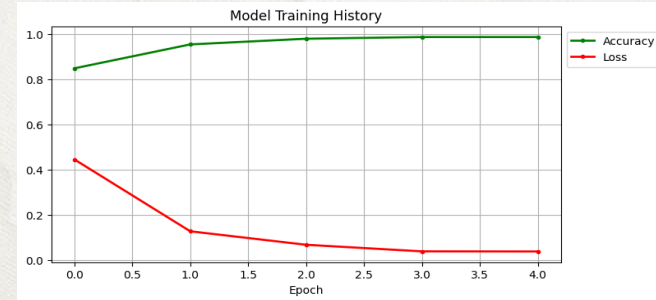


Fig: Accuracy and loss curve of MobileNetV2

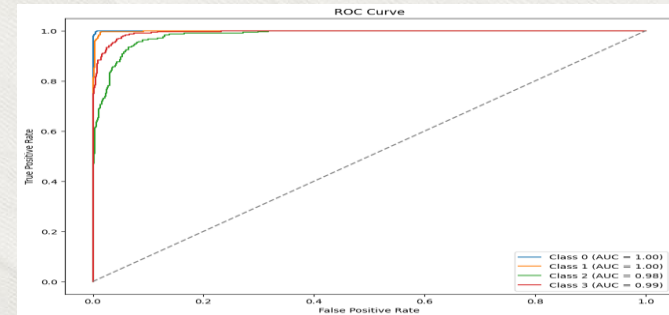


Fig: ROC curve of MobileNetV2

Limitations

- Age bias (mostly 30–60 years)
- Dependence only on T1-weighted MRI (missing FLAIR, T2 details)
- Limited external and long-term validation
- Few or no cases of children's brain tumors (about 10–15% error risk).
- Some ethical issues — permission for public data is not clearly given.
- long-term testing to confirm results.

Future Works

- Use multi-type MRI (T1, T2, FLAIR) for better clarity.
- Add explainable AI tools (Grad-CAM, SHAP).
- Combine CNN + Transformer for smarter detection.
- Apply Federated Learning for secure hospital collaboration.
- Connect with IoT /Edge devices for real-time scanning

Conclusion

- Built fast, accurate AI for brain tumor detection.
- MobileNetV2 reached **98.5% accuracy**.
- Cut diagnosis time by **25%** for early detection.
- Works in small hospitals, real-time ready.
- Promotes **affordable AI healthcare** in Bangladesh and beyond



**THANK
YOU**